

Extracellular vesicle-mediated release of bis(monoacylglycerol)phosphate is regulated by LRRK2 and Glucocerebrosidase activity

Reviewed Preprint

v1 • May 23, 2025

Not revised

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eLife Assessment

This **useful** study presents the potentially interesting concept that LRRK2 regulates cellular BMP levels and their release via extracellular vesicles, with GCase activity further modulating this process in mutant LRRK2-expressing cells. However, the evidence supporting the conclusions remains **incomplete**, and certain statistical analyses are **inadequate**. This work would be of interest to cell biologists working on Parkinson's disease.

<https://doi.org/10.7554/eLife.106330.1.sa2>

Abstract

The endo-lysosomal phospholipid, bis(monoacylglycerol)phosphate (BMP), is aberrantly elevated in the urine of Parkinson's patients with mutations in genes encoding leucine-rich repeat kinase 2 (LRRK2) and glucocerebrosidase (GCase). Because BMP resides on and regulates the biogenesis of endo-lysosomal intraluminal membranes that become extracellular vesicles (EVs) upon release, we hypothesized that elevated urinary BMP may be driven by increased exocytosis of BMP-enriched EVs. To test this hypothesis, we analyzed BMP metabolism and EV-associated release of BMP in wild type (WT) and R1441G LRRK2-expressing mouse embryonic fibroblast (MEF) cells. Using immunofluorescence microscopy and transmission electron microscopy we detected structural alterations in endo-lysosomes and antibody-accessible BMP pool, indicating that mutant LRRK2 affects endolysosomal homeostasis. Biochemical analyses of isolated EV fractions confirmed the effect on endo-

lysosomes by showing an increase in LAMP2-positive EVs in mutant cells, which was partially restored by LRRK2 kinase inhibition but further augmented by GCase inhibition. Using mass spectrometry, we detected an overall increase in total di-22:6-BMP and total di-18:1-BMP in cell lysates from mutant LRRK2 MEFs compared to WT cells. Inhibition of LRRK2 kinase partially restored cellular BMP levels, whereas inhibition of GCase further increased the BMP content. In isolated EVs from LRRK2 mutant cells, LRRK2 inhibition decreased BMP content whereas GCase inhibition tended to increase it. Using metabolic labeling experiments, we demonstrated that the increase in cellular BMP content is not due to an increase in BMP synthesis, even though we observed an increase in BMP synthesizing enzyme, CLN5, in LRRK2 mutant MEFs and patient-derived fibroblasts. Finally, pharmacological modulators of EV release confirmed that BMP release is associated with EV secretion. Together, these results establish LRRK2 as a regulator of BMP levels in cells and its release through EVs, with GCase activity further modulating this process in LRRK2 mutant cells. Mechanistic insights from these studies have implications for the potential use of BMP-positive EVs as a biomarker for Parkinson's disease and associated treatments.

Introduction

Variants in the gene encoding Leucine rich repeat kinase 2 (*LRRK2*) and β glucocerebrosidase (GCase; *GBA1*) together constitute the largest contributors to familial and sporadic Parkinson's disease (PD)^{1,2}. Although G2019S is the most prevalent PD-associated *LRRK2* mutation, R1441G is more penetrant and is associated with an earlier disease onset³⁻⁴. A shared mechanism of pathogenesis among all PD-causing *LRRK2* mutations is increased *LRRK2* kinase activity, which leads to phosphorylation of its cognate Rab GTPase substrates, including Rab8, Rab10 and Rab12, that are key regulators of intracellular vesicle trafficking⁵. A prevailing hypothesis of pathogenesis in *LRRK2* PD cases is that *LRRK2* phosphorylation of Rabs results in defective autophagy and endo-lysosomal homeostasis and defective ciliogenesis in the nigrostriatal circuit^{5,6}.

GBA1 encodes a lysosomal hydrolase, glucocerebrosidase (GCase), essential for glycosphingolipid degradation. Homozygous loss-of-function mutations in *GBA1* cause Gaucher's disease, a lysosomal storage disorder characterized by cytotoxic accumulation of lipid substrates of GCase, glucosylceramide and glucosylsphingosine; heterozygous carriers have a 5-10 fold increased risk of PD^{7,8}. The predominant hypothesis for the pathogenic mechanism of *GBA1* variants is related to aberrant aggregation and clearance of alpha-synuclein in a feed-forward cascade^{9,10}; a hypothesis being tested clinically using GCase activators. Additionally, we and others previously showed that GCase enzymatic function is modulated by *LRRK2* kinase activity^{11,12}. Altogether, it is possible that dysregulation of the endo-lysosomal biology is associated with pathogenic variants of both *LRRK2* and *GBA1*^{6,13}.

Bis(monoacylglycerol)phosphate (BMP, also known as LBPA or Lysobisphosphatidic acid) is an atypical, negatively-charged glycerophospholipid important for maintaining endo-lysosomal homeostasis¹⁴. BMP resides primarily in endo-lysosomal intraluminal vesicles (ILVs) where it participates in multiple processes, including cholesterol egress, exosome biogenesis and lipid catabolism^{14,15}. Of note, BMP acts together with Saposin-C as enzymatic cofactors of GCase^{16,17}. Multiple BMP isomers are present in cells, based on differences in their fatty acyl chains and their positions relative to the sn-2 and sn-3 carbon atoms on each of its two glycerol backbones^{14,15}; the sn-2:sn-2' isomer is the proposed active form^{18,19}. Dioleoyl-BMP (di-18:1-BMP) is the predominant form in numerous cell lines analyzed²⁰⁻²³ and didocosahexaenoyl-BMP (di-22:6-BMP) is one of the most abundant species in the brain^{24,25}. Fatty acyl composition is thought to influence BMP biophysical properties and function²⁶.

We and others recently reported elevated levels of total di-18:1-BMP and total di-22:6-BMP species in the urine of carriers of PD-associated *LRRK2* and *GBA1* mutations^{27–29}. Although we did not observe a correlation between urinary BMP and PD progression²⁸, these studies underscore BMP's utility as a patient enrichment and target modulation biomarker in therapeutic trials³⁰. Based on our previous findings, in the present study we focused on these two BMP species.

Given the specific intracellular localization of BMP (in ILVs; exosomes when released) and its proposed roles in exosome biogenesis¹⁹, we hypothesized that its increased levels in PD biofluids may be a consequence of an increase in the secretion of BMP containing extracellular vesicle (EV). To gain mechanistic insight into the regulation and biological significance of BMP release, we monitored BMP release in EVs and performed kinetic analyses of BMP biosynthesis in mouse embryonic fibroblasts (MEFs) from wild-type (WT) or R1441G *LRRK2* knock-in mice. In addition, we examined the regulation of BMP by a *LRRK2* kinase inhibitor, MLI-2, and a GCase inhibitor, conduritol β -epoxide (CBE), to gain insights into the role of BMP in potential therapeutic vs. pathophysiological responses, respectively. Our data indicate that *LRRK2* and GCase activities modulate its release in EVs.

Results

Analysis of BMP-positive endo-lysosomes in R1441G *LRRK2* mouse embryonic fibroblasts

We first performed immunofluorescence microscopy of BMP-positive endo-lysosomes in both WT and R1441G *LRRK2* knock-in MEFs. The overall antibody-accessible BMP pool was significantly decreased in mutant *LRRK2*-expressing cells compared to WT (**Figure 1A, B**). In contrast, endo-lysosomal LAMP2 fluorescence was higher in mutant *LRRK2* MEFs (**Figure 1A, C**). Previous studies found structural defects in endo-lysosomes of different cell types harboring pathogenic *LRRK2* mutations^{31–33}. Since BMP is specifically enriched in intra-luminal vesicles (ILVs) in the endo-lysosomes where it regulates important catabolic reactions, we examined multivesicular endosome (MVE) morphology of these cells. In agreement with previous findings³³, transmission electron microscopy revealed decreased MVE area in R1441G *LRRK2* MEFs (average area = $0.11 \mu\text{m}^2$) compared to WT MEFs (average area = $0.27 \mu\text{m}^2$) (**Figure 1D, E**). In addition, the overall endo-lysosomal ILV number was diminished in the R1441G *LRRK2* cell line (**Figure 1F**). These data suggest that the pathogenic *LRRK2* mutation alters BMP-positive endo-lysosome morphology and ILV content.

LRRK2 and GCase activities modulate extracellular vesicle release

Given the previously described roles of BMP in ILV/exosome biogenesis, we investigated whether the alterations observed in antibody-accessible BMP and MVE ILV number in *LRRK2* mutant MEFs could be explained by changes in EV release. We isolated and characterized EVs from both WT and R1441G MEFs and also assessed the effects of *LRRK2* and GCase pharmacological inhibitors, MLI-2 and CBE, respectively on EV content and number. Inhibition of *LRRK2* kinase and GCase enzymatic activities was confirmed, respectively, by monitoring phospho-Rab10 levels in whole cell lysates and performing a fluorescence-based GCase activity assay in cells (**Figure 2A, B**). Consistent with our immunofluorescence data, upregulation of LAMP2 was observed in R1441G *LRRK2* whole cell lysates (WCL; **Figure 2A, C, D** and **E**), as previously reported in *LRRK2* G2019S knock-in mouse brain³⁴. MLI-2 treatment for 48 hours partially reversed this phenotype (**Figure 2A, C, E**), suggesting that aberrant *LRRK2* kinase activity influences endo-lysosomal homeostasis.

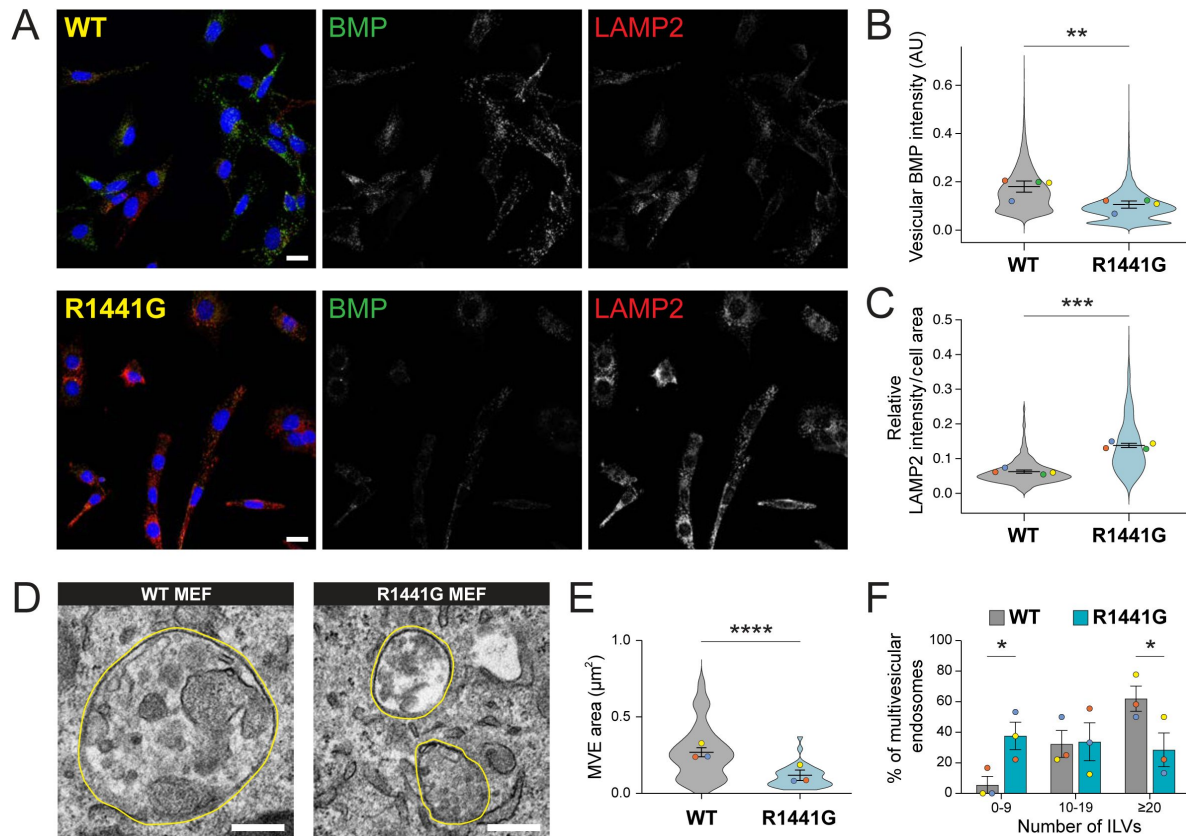


Figure 1.

Alterations of antibody-accessible BMP and endolysosomal morphology in LRRK2 R1441G-MEF cells

(A) Confocal microscopy of endogenous BMP (green) and LAMP2 (red) immunofluorescence in WT and R1441G LRRK2 mutant MEF cells. Scale bar: 20 μm . (B-C) Quantification of vesicular BMP intensity (B) and LAMP2 relative intensity (C) per cell area. Colored dots represent mean value from 4 independent experiments and violin plots show the distribution of individual cell data. Significance determined by two-tailed paired *t* test ***p*<0.01, ****p*<0.001. (D) Representative transmission electron microscopy (TEM) images of Multivesicular endosomes (MVE) from WT and R1441G LRRK2 mutant MEF cells. MVB periphery highlighted in yellow. Scale bar: 250nm. (E) MVE area (μm^2) quantification in WT and R1441G LRRK2 mutant cells. Colored dots represent mean values from 3 independent experiments and violin plots show the distribution of individual cell data (35-45 cells/group). (F) Quantification of Intraluminal Vesicles (ILVs) per MVE in WT and R1441G mutant MEF cells. The number of ILVs per MVE are binned in three groups and plotted as a percentage of MVE from the total population of each experiment independently. Data from 3 independent experiments (mean $\pm$ SEM). Significance determined by two-tailed unpaired *t* test (E) and ordinary two-way ANOVA, uncorrected Fisher's LSD (F) **p*<0.05, *****p*<0.0001.

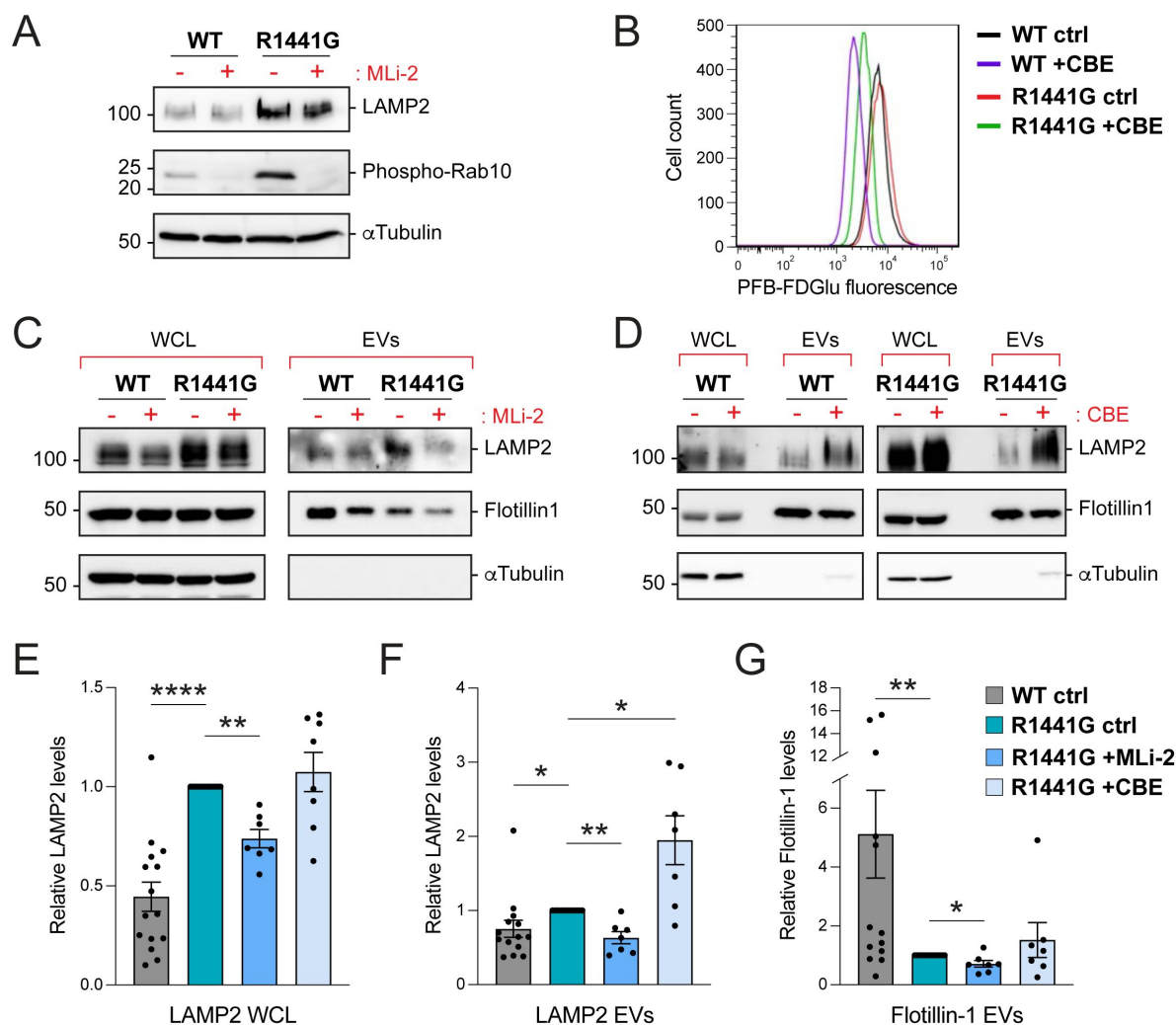


Figure 2.

LRRK2 and GCase activities modulate extracellular vesicle production

(A) Whole cell lysates from WT and R1441G mutant MEF cells treated with 200nM MLI-2 for 24h were analyzed by western blotting. Representative images of LAMP2, phospho-Rab10 and α-Tubulin levels are shown. Molecular weight marker mobility is shown in kDa. **(B)** Flow cytometry measurement of GCase activity using PFB-FDGlu fluorescent GCase substrate in WT and R1441G LRRK2 mutant MEF cells treated with 300μM CBE for 24h. **(C-D)** Whole cell lysates (WCL) and isolated extracellular vesicles (EVs) from WT and R1441G LRRK2 mutant MEF cells treated with 200nM MLI-2 **(C)** or 300μM CBE **(D)** for 48h were analyzed by western blotting. Representative images of LAMP2, Flotillin-1 and α-Tubulin levels are shown. Molecular weight marker mobility is shown in kDa. **(E-G)** Quantification of LAMP2 and Flotillin-1 levels relative to R1441G LRRK2 MEF cells in whole cell lysates (WCL). **(E)** and isolated EVs **(F-G)**. Data from 6-9 independent experiments (mean ± SEM). Significance determined by two-tailed paired *t* test compared to R1441G LRRK2 control **p*<0.05, ***p*<0.01, *****p*<0.0001.

We next analyzed the profiles of isolated EV fractions from both wild-type (WT) and R1441G LRRK2 mutant MEFs. Both LAMP2 and Flotillin-1 were assessed to explore alterations in release of EV subpopulations; LAMP2 is enriched in ILV-derived EVs while Flotillin-1 is also seen in plasma membrane-derived ectosomes that reflect outward budding of the plasma membrane³⁵. Biochemical analysis revealed an elevation of LAMP2, but not Flotillin-1, in EVs derived from R1441G LRRK2 MEFs compared to those from WT MEFs (**Figure 2C, F, G**). Upon MLI-2 treatment, a modest but significant decrease in both EV-associated protein markers was observed in isolated EV fractions from mutant LRRK2 MEFs (**Figure 2C, F, G**). In contrast, inhibition of GCase activity in mutant LRRK2 cells led to an increase in LAMP2, but not Flotillin-1, in isolated EVs compared to those derived from WT MEFs (**Figure 2C, F, G**). Finally, analysis of WT cell-derived EVs revealed no major differences in EV marker levels between untreated (control) and MLI-2-or CBE-treated conditions (**Supplemental Figure 1A-C**).

To complement these data, we conducted nanoparticle tracking analysis (NTA). Isolated EV fractions from WT and R1441G LRRK2 cells exhibited comparable particle size distributions (**Supplemental Figure 1D**). Treatments with MLI-2 and CBE yielded measurable quantitative changes in EV concentrations that did not reach statistical significance (**Supplemental Figure 1E**), potentially reflecting the inherent variability of NTA due to its inability to distinguish EVs from non-vesicular particles^{37,38}. For the R1441G MEF cells, MLI-2 decreased EV concentration while CBE increased EV particles per ml, in agreement with the effects observed in our biochemical analysis.

Altogether these results suggest that EV secretion is influenced by LRRK2 kinase and GCase hydrolase activities; while MLI-2 treatment decreases EV release, CBE increases it.

Targeted BMP lipid analysis in intact cells and extracellular vesicles

Ultra-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) was used to measure quantitatively the impact of LRRK2 and GCase activities on BMP isoform abundance in cells and isolated EV fractions. Remarkably, an overall increase in total BMP isoforms was detected in mutant LRRK2 MEF cell lysates (**Figure 3A-D**), with di-22:6-BMP and di-18:1-BMP as the major species. It is important to note that mass spectrometry-based methods detect total BMP content while antibody staining (**Figure 1A, B**) only detects so-called “antibody-accessible” BMP³⁹. The mass spectrometry-based increase in total BMP in LRRK2 mutant cells indicates that this pool is less antibody-accessible than that present in wild type cells. Alternately, the anti-BMP antibody may be less specific and detect other analytes.

In R1441G cells, LRRK2 inhibition for 48 hours with MLI-2 decreased total di-22:6-BMP and total di-18:1-BMP levels by ~20% (**Figure 3A** and **B**). Conversely, inhibition of GCase increased total BMP levels, reaching significance for di-22:6-BMP levels compared with untreated, R1441G LRRK2 cells (**Figure 3C**). No statistically significant differences in intracellular BMP levels were observed in WT LRRK2 MEFs upon LRRK2 or GCase inhibition (**Supplemental Figure 1D, E**), suggesting a dominant role of mutant LRRK2 activity in BMP regulation. These data suggest that there are LRRK2-independent clonal differences leading to differences in basal BMP content between WT and mutant MEF cells; however, such clonal variation does not impact the effect of MLI-2 or CBE treatment in R1441G cells.

Analysis of isolated EV fractions detected only the di-18:1-BMP isoform, with no evidence of di-22:6-BMP. Although a trend toward higher di-18:1-BMP levels was observed in EVs derived from WT cells compared to those from mutant LRRK2 cells, this difference was not statistically significant (**Figure 3E**). Treatment of LRRK2 R1441G MEFs with MLI-2 resulted in a significant partial decrease in EV-associated total BMP (**Figure 3E**), consistent with previous observations in non-human primates and PD mouse models that showed decreased extracellular urinary BMP

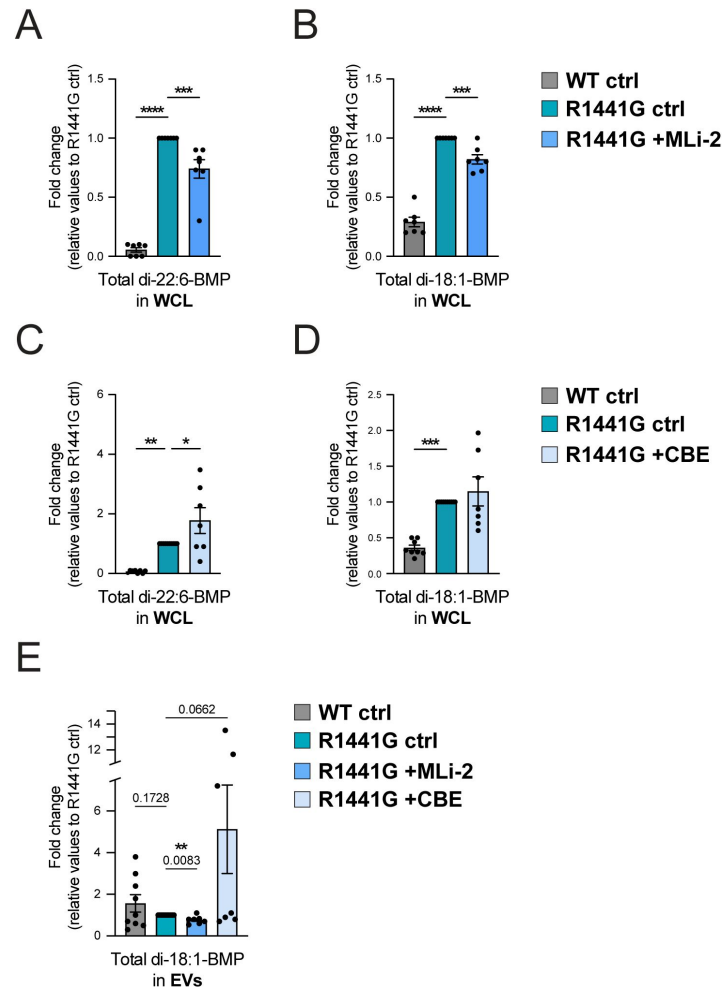


Figure 3.

Targeted lipid pathway analysis of BMP abundance in cellular and isolated EV fractions.

(A-D) UPLC-MS/MS determination of BMP isoforms normalized to protein content from cells treated with 200nM MLI-2 (A-B) or 300μM CBE (C-D) for 48h. Data shown as fold change relative to R1441G LRRK2 control MEF cells. Only BMP isoforms that were detected are shown. (E) UPLC-MS/MS determination of BMP isoforms normalized to protein content in EVs isolated from cells treated with 200nM MLI-2 or 300μM CBE for 48h. Only BMP isoforms that were detected are shown. Data from 3-6 independent experiments (mean ± SEM). Significance determined by ordinary one-way ANOVA, uncorrected Fisher's LSD (A-D) and one-way ANOVA with the Geisser-Greenhouse correction, uncorrected Fisher's LSD (E) *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

upon LRRK2 kinase pharmacological inhibition^{40–42}. In contrast, inhibition of GCase activity yielded an opposite, albeit not significant, trend (**Figure 3E**). These latter findings may be explained by the observation that GCase inhibition by CBE was less pronounced in LRRK2 R1441G cells compared to WT cells under identical concentration and treatment duration conditions (**Figure 2B**). Finally, no significant differences in EV-associated BMP abundance between untreated (control) and MLI-2-or CBE-treated WT LRRK2 MEFs were observed (**Supplemental Figure 1F**).

In addition to analyzing BMP, we also examined GCase lipid substrates in both cells and isolated EVs. Targeted quantification of sphingolipid species revealed elevated GCase substrate abundance in both CBE-treated WT and mutant LRRK2 cells, validating the treatment paradigm used in this study. Interestingly, CBE-mediated accumulation of GCase substrates was more evident in R1441G LRRK2 than in WT cells (**Supplemental Figure 2A**). This suggests that, despite a less pronounced direct effect on GCase activity (**Figure 2B**), the R1441G mutation might contribute to broader endo-lysosome dysfunction, which could alter the dynamics of substrate accumulation. On the other hand, analysis of isolated EV fractions revealed lower levels of glucosylceramide, galactosylceramide, and glucosylsphingosine in EVs from R1441G LRRK2 cells compared to those from WT MEFs, but no significant differences were detected between control or CBE-treated conditions independent of LRRK2 mutation status (except for glucosylsphingosine, which showed an increase in WT-EVs upon GCase inhibition; **Supplemental Figure 2B**).

Altogether, given that BMP is specifically enriched in ILVs (which become exosomes upon release), the data presented above support our biochemical analysis (**Figure 2C, D, F**) and suggest a role for LRRK2 and GCase in modulating BMP release in association with LAMP2-positive exosomes from MEF cells.

BMP biosynthesis is not influenced by LRRK2 kinase and GCase activities

To rule out the possibility that differences in cellular and EV-associated BMP levels following MLI-2 or CBE treatment of R1441G cells were caused by alterations in BMP metabolism rather than changes in membrane trafficking and EV release, we performed metabolic labeling experiments using heavy (H) isotope-labeled 22:6 and 18:1 fatty acids as BMP precursors.

Cells were pulsed for 25_min with heavy isotope BMP precursors and then washed and chased for different time points (**Figure 4A**). UPLC-MS/MS analysis allowed us to differentiate unlabeled versus semi-labeled (H,L'; only one heavy isotope-labeled fatty acid chain) or fully-labeled (H,H'; both fatty acid chains heavy isotope-labeled) BMP species (**Figure 4A**). Initial experiments were performed over 48h using mutant LRRK2 cells under control and LRRK2-or GCase-inhibition conditions; untreated WT cells were included for comparison. Strikingly, even at time 0 and throughout the length of the experiment, R1441G LRRK2 cells displayed substantially higher H,L'- and H,H'-22:6-BMP species than WT cells (left and middle plots in **Figure 4B**). H,L', but not H,H' 18:1-BMP was detected, which appeared to increase to a greater extent in R1441G LRRK2 cells at least during the initial 8h of the experiment (right plot, **Figure 4B**). Despite these differences between WT and mutant LRRK2 cells, neither inhibition of LRRK2 or GCase had a major impact in the kinetics of BMP synthesis or catabolism (**Figure 4B**). To better resolve initial BMP synthesis rates, we reduced the pulse labeling time and chase time to 60 min. In this set of experiments, only isotope-labeled 22:6-BMP species were detected (**Figure 4C**), possibly reflecting a preference for 22:6 BMP synthesis by MEFs. As before, even at time 0 and throughout the 60 min duration of the assay, 22:6-BMP levels were consistently higher in R1441G LRRK2 cells compared with WT MEFs. Again, no overall rate differences were seen between untreated and MLI-2-or CBE-treated cells. These experiments support the hypothesis that rather than altering BMP metabolic rates, LRRK2 and GCase activities influence EV-mediated BMP release.

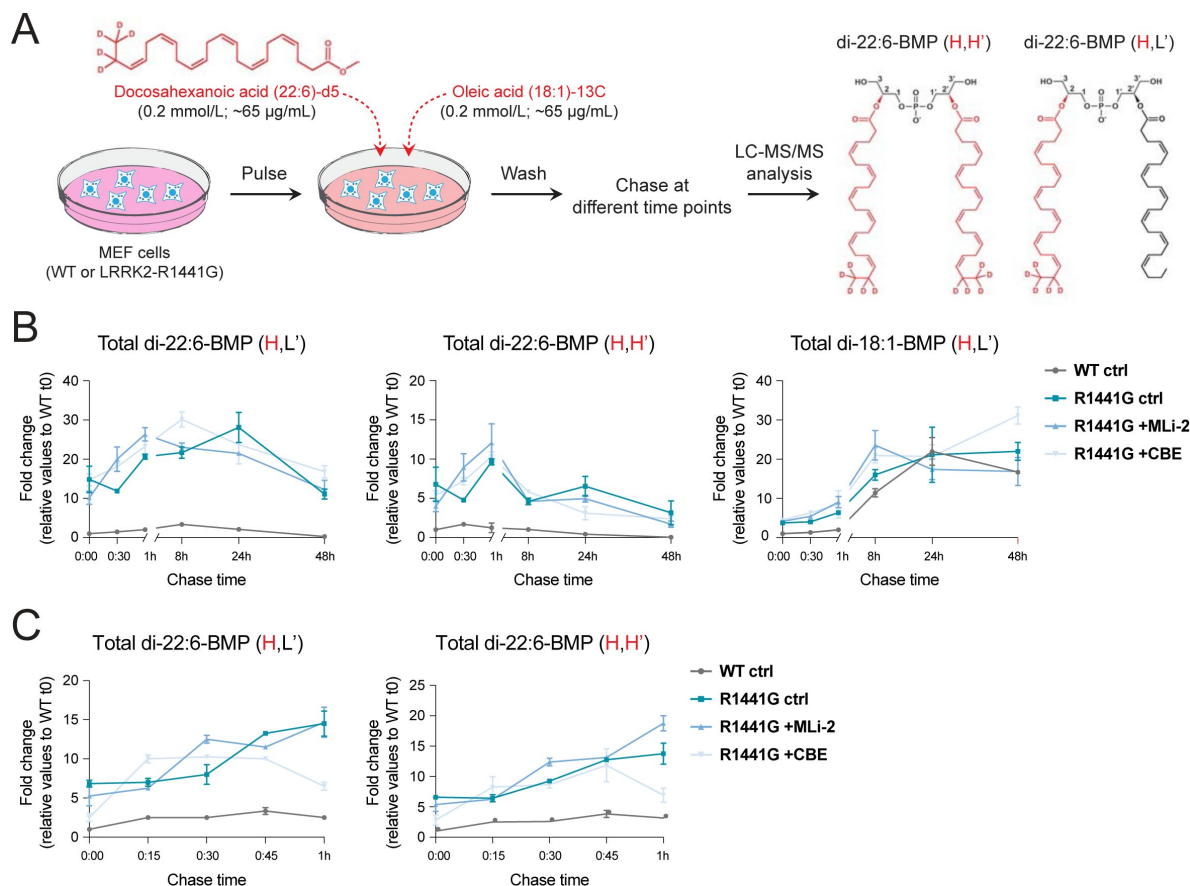


Figure 4.

Inhibition of LRRK2 or GCase activities does not significantly impact BMP biosynthetic and catabolic rates

(A) Schematic representation of the BMP metabolic labeling protocol with deuterated docosahexaenoic acid and ^{13}C -labeled oleic acid. WT and R1441G mutant MEF cells were incubated with a pulse of DHA-d5 / OA- ^{13}C for 20min, followed by a chase for different times. Cells were then collected for subsequent BMP lipidomic analysis. Structures of unlabeled (L) and isotope-labeled (H) fatty acids are shown in black or red, respectively. (B-D) UPLC-MS/MS determination of BMP isoforms normalized to protein content from WT MEF cells and R1441G LRRK2 mutant MEF cells $\pm$ MLI-2 (200nM) or CBE (300µM). Long (B-C) and short (D) chase time points shown as fold change relative to WT control MEF cells time 0. Only BMP isoforms that were detected are shown. Data from 3 replicated experiments (mean $\pm$ SEM).

Increased BMP synthase protein expression in mutant LRRK2 cells

To further investigate the biological significance of BMP upregulation observed in mutant LRRK2 MEF cells (**Figure 3A-D**; **Figure 4B, C**), we examined the expression of CLN5, a key lysosomal enzyme in the BMP biosynthetic pathway^{43,44}. Interestingly, biochemical analysis of total cell lysates revealed a significant fold-change increase in CLN5 protein levels in R1441G LRRK2 MEFs relative to WT cells (**Figure 5A**). Notably, 16 hour treatment with MLI-2 reduced CLN5 levels to a comparable fold-change extent (**Figure 5B**). To validate these findings in a human and disease-relevant context, we examined CLN5 protein expression in cultured fibroblasts derived from Parkinson's disease (PD) patients carrying the G2019S LRRK2 mutation. Consistent with our observations in MEF cells, CLN5 protein levels were reduced following 16 hr MLI-2 treatment in a concentration-dependent manner (**Figure 5C**). Notably, the maximal reduction was observed at the same 200nM MLI-2 concentration used in MEF cell experiments, a dose that also did not induce observable cytotoxic effects in human fibroblasts. Similar results were obtained in patient-derived fibroblasts harboring the R1441G LRRK2 mutation (**Figure 5D**). Altogether, these data suggest that LRRK2 kinase activity may regulate CLN5 protein expression. The upregulation of CLN5 may be due to an overall upregulation of lysosomal enzymes as LAMP2 levels were also increased (**Figure 2A, C, E**). Moreover, the observed baseline differences in BMP, as documented in the flux study above (**Figure 4B** and **C**), could result from CLN5 upregulation. The lack of significant changes in the BMP synthesis rate (**Figure 4B** and **C**) suggests either a limitation in substrate availability or that CLN5 is operating at maximal capacity.

BMP release is EV-mediated

Our data suggest that BMP is exocytosed in association with EVs and that LRRK2 and GCase activities modulate BMP secretion. To determine the magnitude of BMP release via EV secretion, we assessed the impact of pharmacological modulators of EV release. Treatment of WT MEFs with GW4869, a selective type 2-neutral sphingomyelinase inhibitor, decreased EV release as monitored by reduced levels of LAMP2 and Flotillin-1 in EV fractions (**Figure 6A**). Under these conditions, we observed a parallel decrease in exosomal BMP (**Figure 6A, B**). In contrast, enhancing EV release with bafilomycin A1 (B-A1), a pharmacological inhibitor of the endo-lysosomal proton pump V-ATPase that dramatically boosts EV exocytosis^{35,45}, resulted in the opposite trend (**Figure 6A, C**): EV markers and BMP levels increased. Biochemical analysis of EV release in R1441G LRRK2 cells showed similar behavior as in WT MEFs upon treatment with these agents (**Figure 6A** and **Supplemental Figure 3**). GW4869 inhibition of EV release increased cellular BMP content (**Figure 5D**), while enhanced EV release upon B-A1 treatment diminished intracellular total di-22:6-BMP and di-18:1-BMP (**Figure 5E**). Altogether, these results strongly support the notion that BMP is released by EV-mediated MVE exocytosis.

Discussion

Here we have shown that: (a) hyperactive LRRK2 kinase upregulates mass spectrometry-determined BMP levels and the extracellular vesicles (EVs) that release BMP in MEF cells. (b) LRRK2 and GCase pharmacological inhibitors, MLI-2 and CBE, modulate BMP abundance in cells and isolated exosomal fractions in opposite directions without interfering with the kinetics of BMP biosynthetic or catabolic rates. While PD-associated mutations in *LRRK2* increase kinase activity, *GBA1* pathogenic variants are associated with decreased GCase enzymatic function. Therefore, MLI-2 treatment represents a potential therapeutic rescue, whereas CBE phenocopies pathogenic conditions. (c) Although the hyperactive kinase mutation is associated with an increase in the rate of BMP synthesis, neither MLI-2 or CBE affected BMP synthesis in the MEFs. (d) The expression of CLN5, a BMP synthase⁴³, is upregulated in mouse and patient-derived fibroblasts with PD-

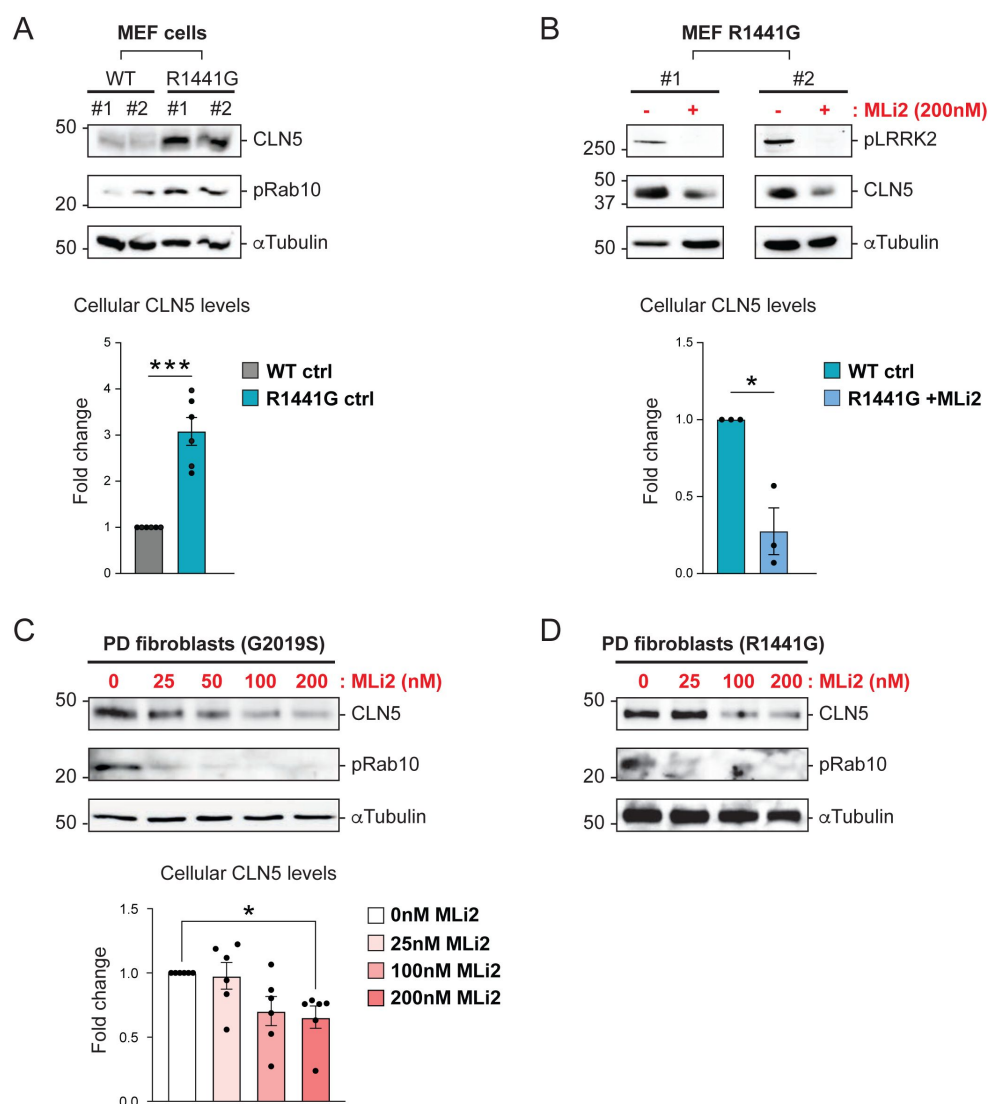


Figure 5.

LRRK2 activity modulates CLN5 expression levels

A) Whole cell lysates from WT and R1441G LRRK2 MEF cells were analyzed by western blotting. Representative immunoblots of CLN5, phospho-Rab10 (pRab10) and α-Tubulin are shown from two (#1 and #2) out of six independent experiments. Molecular weight marker mobility is shown in kDa. Plot at the bottom shows quantification of CLN5 immunoblot levels relative to WT and R1441G LRRK2 MEF cells. **B)** Whole cell lysates from R1441G LRRK2 MEF cells treated with 200nM MLI-2 for 24h were analyzed by western blotting. Representative immunoblots of phospho-LRRK2 (pLRRK2), CLN5 and α-Tubulin levels are shown from two (#1 and #2) out of three independent experiments. Molecular weight marker mobility is shown in kDa. Plot at the bottom shows quantification of CLN5 immunoblot levels in whole cell lysates of R1441G LRRK2 MEF cells untreated or MLI-2-treated. **C)** Whole cell lysates from G2019S LRRK2 patient-derived fibroblasts treated with indicated increasing MLI-2 concentrations for 24h were analyzed by western blotting. Immunoblots of CLN5, phospho-Rab10 (pRab10) and α-Tubulin levels are shown from one representative experiment (n=6 per condition, obtained from two independent replicate experiments using fibroblast cell lines derived from three different patients). Molecular weight marker mobility is shown in kDa. Plot at the bottom shows quantification of CLN5 immunoblot levels relative to G2019S LRRK2 patient-derived fibroblasts treated with MLI-2 at the indicated concentrations. **D)** Whole cell lysates from R1441G LRRK2 patient-derived fibroblasts treated with indicated increasing MLI-2 concentrations for 24h were analyzed by western blotting. Immunoblots of CLN5, phospho-Rab10 (pRab10) and α-Tubulin levels are shown from one representative experiment. In **A)** and **B)** significance determined by two-tailed, unpaired *t* test; significance in **C)** determined by Dunnett's One-way ANOVA test; **p*<0.05, ****p*<0.001.

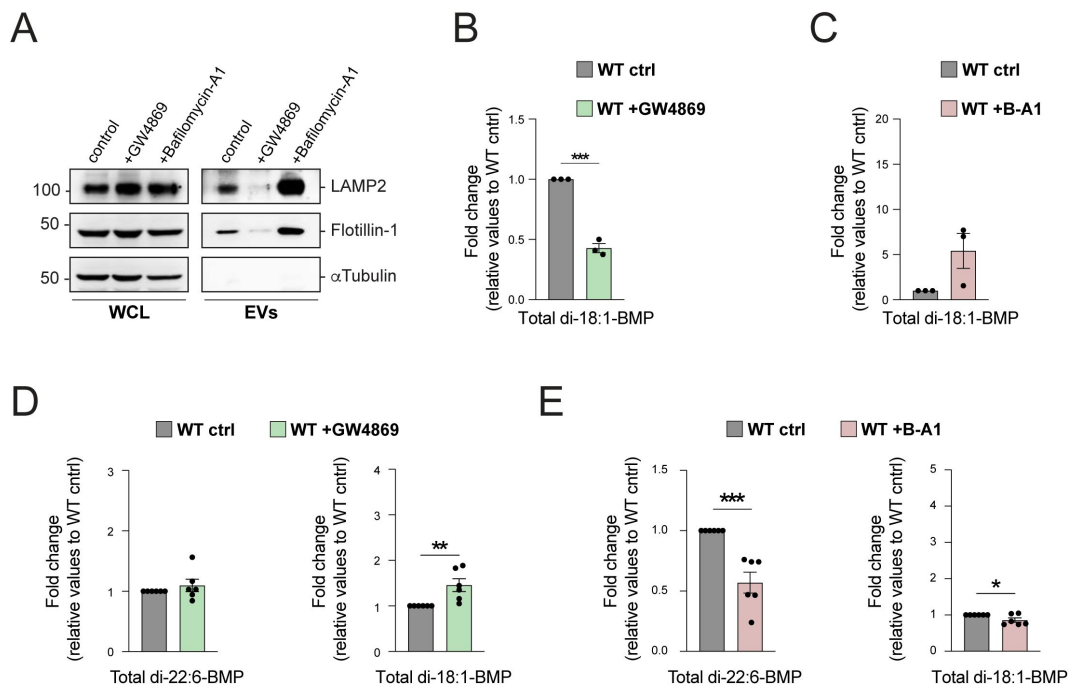


Figure 6.

Pharmacological modulation of EV-mediated BMP exocytosis

(A) Whole cell lysates (WCL) and isolated EVs from WT MEF cells treated with 10μM GW4869 or 10nM bafilomycin-A1 (B-A1) for 24h were analyzed by western blotting. Representative immunoblots of LAMP2, Flotillin-1 and α-Tubulin are shown. Molecular weight marker mobility is shown in kDa. **(B-C)** UPLC-MS/MS determination of BMP isoforms normalized to protein content from WT MEF cells treated with 10nM B-A1 **(B)** or 10μM GW4869 **(C)** for 24h. Data shown as fold change relative to WT control MEF cells. Only BMP isoforms that were detected are shown. Data from 6 independent experiments (mean ± SEM). Significance determined by two-tailed unpaired *t* test **p*<0.05, ***p*<0.01, ****p*<0.001, *****p*<0.0001. **(D-E)** Quantitation of BMP isoforms normalized to protein content in EVs isolated from MEF WT cells treated with 10nM B-A1 **(D)** or 10μM GW4869 **(E)** for 24h. Data shown as fold change relative to WT control MEF cells. Only BMP isoforms that were detected are shown. Data from 3 independent experiments (mean ± SEM). Significance determined by two-tailed unpaired *t* test **p*<0.05, ***p*<0.01, ****p*<0.001.

associated LRRK2 mutations, consistent with the flux data indicating elevated overall BMP synthesis. (e) Finally, BMP release was modulated by pharmacological agents known to modulate EV secretion. Together, these data indicate that the previously reported increase in urinary BMP levels in LRRK2 mutation carriers indicate dysregulated exocytosis of BMP-containing EVs.

Pioneering studies by Gruenberg and colleagues underscore the importance of BMP in endo-lysosome homeostatic regulation and generation of endo-lysosomal ILVs^{14,18,19} that become exosomes upon release. Subsequent studies revealed that cellular levels of this atypical phospholipid are invariably altered in many neurodegenerative disorders characterized by endo-lysosome dysfunction²⁴. Recently, key lysosomal enzymes involved in distinct steps of the BMP biosynthetic pathway were identified, including CLN5 which plays a critical role in this process^{43,44}. Our microscopy analysis revealed decreased antibody-accessible BMP levels in R1441G LRRK2 cells. In contrast, our targeted lipid pathway analysis measurements consistently showed higher total BMP levels in R1441G LRRK2 cells compared with WT cells. As we have reported previously³⁹, BMP antibody detects only a sub-pool of “accessible” BMP while mass spectrometry detects all pools. Alternately, the antibody may also detect non-BMP analytes. Given the essential roles of BMP in endo-lysosomal catabolism, our immunofluorescence data predict that LRRK2 mutant cells have defective degradative capacity, consistent with recent reports^{31,46}. In addition, we found a concomitant upregulation of LAMP2 and the BMP-synthesizing enzyme CLN5, suggesting that LRRK2 may participate in endo-lysosome biogenesis regulation. Indeed, in macrophages and microglia, LRRK2 regulates the levels of multiple lysosomal proteins by inhibiting TFEB and MITF⁴⁶, transcription factors that activate lysosome-related gene expression by binding to coordinated lysosomal expression and regulation (CLEAR) elements⁴⁷. Although the promoter region of CLN5 contains a potential TFEB binding site⁴⁷, formal evidence for TFEB-mediated transcription of CLN5 is still lacking. The increase in CLN5 and LAMP2 protein levels may also reflect lysosomal stress⁴⁷. Interestingly, MLI2 restored CLN5 levels in both murine MEFs and human fibroblasts, further providing support that BMP is a pharmacodynamic biomarker of not just target modulation but also a potential disease-relevant lysosomal pathway modulation. Our findings of CLN5 upregulation in mutant LRRK2 MEFs, validated by similar results in patient-derived fibroblasts, further establish MEFs as a relevant cellular model system for studying LRRK2 kinase activity and its therapeutic targeting^{48–51}.

We and others reported that BMP is aberrantly high in urine derived from LRRK2 and GBA mutation-carriers^{27–29}. These results have also been reproduced in animal models, in which urinary BMP levels decrease upon administration of MLI-2 and other LRRK2 kinase inhibitors^{40–42}. The present study complements these previous observations by providing evidence of EV-mediated BMP release that can be regulated by LRRK2 and GCase enzymatic activity. Consistent with this model, recent studies have shown that aberrant LRRK2 and GCase activities influence EV secretion in cellular and animal models of PD, including human patients^{52–54}. Moreover, our data showing reduced EV release in mutant LRRK2 MEFs treated with MLI-2 is consistent with previous observations that LRRK2 kinase inhibition leads to surfactant accumulation in type 2 pneumocytes in the lung, likely as a consequence of impaired lysosome-related organelle exocytosis⁴¹.

BMP is enriched in urinary EVs⁵⁵. However, other studies did not detect BMP enrichment in isolated EV fractions and concluded that BMP may be present in ILVs of endo-lysosomal subpopulations devoted to degradative activities rather than exocytic events^{56,57}, as previously proposed⁵⁸. Using our UPLC-MS/MS methodology, extracellular BMP can be efficiently detected in the urine of healthy and mutant LRRK2/GCase carriers^{27–29}, and additionally, in EV fractions derived from both WT and R1441G LRRK2 MEFs (this study). How does LRRK2 influence release of BMP-positive EVs (BMP-EVs)? A subset of Rab proteins that are master regulators of intracellular membrane trafficking pathways are important LRRK2 substrates⁵. Some of these Rabs include Rab10, Rab12 and Rab35, and have been shown to play regulatory roles in endo-lysosomal exocytosis and EV release^{59,60}. We thus speculate that

phosphorylation of one or several of these Rabs may lead to enhanced BMP release. Indeed, G2019S LRRK2--mediated Rab35 phosphorylation has been proposed to promote EV-mediated α -synuclein release and propagation between cells^{61,62}. It is likely that aberrant LRRK2 and GCase activities trigger endo-lysosomal exocytosis, and associated release of BMP-containing EVs, via a so-called clearance pathway activated due to accumulation of cytotoxic endo-lysosomal substrates^{63,64}. The contribution of these phosphorylated Rabs to EV-mediated BMP release will be of interest for future work.

In summary, this work provides evidence that LRRK2 and GCase pathological activities upregulate EV-mediated BMP release without altering its metabolism in cells. The observed changes induced by MLI-2 highlight the potential of LRRK2 inhibition as a therapeutic strategy to restore endo-lysosomal homeostasis by reducing BMP production and aberrant EV release. What is the clinical significance of elevated urinary BMP levels in PD patients? Inhibition of LRRK2 improves endo-lysosomal function⁴², consistent with increased LRRK2 and decreased GCase activities associated with pathogenic variants which worsen endo-lysosomal homeostasis^{6,13}; this may eventually lead to endo-lysosomal exocytosis followed by EV-mediated BMP release. We speculate that BMP-EVs may harbor a distinct molecular repertoire that could perhaps inform more precisely disease progression or pathobiology. Future investigations will be needed to study the contribution of dysfunctional LRRK2 and GCase in exocytosis of specific BMP-positive endo-lysosomal subpopulations and to further characterize the role of BMP-EVs in the context of disease pathophysiology but also as a PD diagnostic tool.

Materials and methods

Cell culture, antibodies and other reagents

Wild type (WT) and R1441G LRRK2 mutant MEF cells were grown in Dulbecco's modified Eagle's media (DMEM) containing 10% fetal bovine serum, 2 mM L-glutamine, and penicillin (100 U/ml)/streptomycin (100 mg/ml). Cell lines were cultured at 37°C with 5% CO₂. Extracellular vesicle-free media was prepared by overnight ultracentrifugation of DMEM or RPMI supplemented with 10% FBS at 100,000 X g in a SW32Ti rotor. For bafilomycin-A1 (Sigma-Aldrich; catalog no. B1793) and GW4869 (SelleckChem; catalog no. S7609), the drug was added to cell media at 10 nM and 10 μ M respectively and cells were cultured for 24 h before exosome collection. For MLI-2 (Tocris Bioscience; catalog no. 5756) and Condurotol B Epoxide (MERK; catalog no. 234599), the drug was added to cell media at 200nM and 300 μ M respectively and cells were cultured for 48 h before exosome isolation. 0.167mM Oleic acid (Cambridge Isotope Laboratories; catalog no. DLM-10012-0.001) and Docosahexaenoic acid (Cambridge Isotope Laboratories; catalog no. DLM-10012-0.001) were conjugated with 0.0278mM fatty acid-free BSA (Sigma-Aldrich; A8806) at a 1:6 molar ratio. Primary antibodies diluted in PBS with 1% BSA (for immunofluorescence) or 5% skim milk (for immunoblotting) were mouse anti-lysobisphosphatidic acid (anti-BMP) clone 6C4 1:1000 (EMD Millipore; catalog no. MABT837), rat monoclonal anti-mouse LAMP2 1:1000 (Developmental Studies Hybridoma Bank, catalog no. GL2A7), mouse anti-Flotillin-1 1:1000 (BD Biosciences; catalog no. 610821), rabbit anti-phospho-Rab10 1:1000 (Abcam; catalog no. ab230261), mouse anti-alpha tubulin 1:10000 (SigmaAldrich; catalog no. T5168), rabbit anti-CLN5 1:000 (Abcam; catalog no. ab170899), rabbit anti-phosphoLRRK2 1:000 (Abcam; catalog no. ab133450).

Immunofluorescence

Wild type (WT) and R1441G LRRK2 mutant MEF cells were fixed with 3.7% (v/v) paraformaldehyde for 15 min, washed and permeabilized/blocked with 0.1% saponin/1% BSA-PBS before staining with anti-LAMP2 and anti-BMP antibodies followed by goat anti-rat AlexaFluor555 and donkey anti-mouse AlexaFluor488. Nuclei were stained using 0.1 mg/ml DAPI (SigmaAldrich; catalog no. D9542). Coverslips were mounted on glass slides with Mowiol. Microscopy images were acquired using a Zeiss LSM880 laser scanning spectral confocal microscope (Carl Zeiss) equipped with an

Axio Observer 7 inverted microscope, a blue diode (405 nm), argon (488 nm), diode-pumped solid-state (561 nm), and HeNe (633 nm) lasers, and a Plan Apochromat 63×/NA1.4 oil-immersion objective lens. DAPI, Alexa Fluor 488, Alexa Fluor 555 were acquired sequentially using 405-, 488- and 561-nm laser lines; Acousto-Optical Beam Splitter as a beam splitter; and emission detection ranges 415–480 nm, 500–550 nm, 571–625 nm, respectively. The confocal pinhole was set at 1 Airy unit. All images were acquired in a 1,024 × 1,024-pixel format. BMP and LAMP2 intensities were quantified using CellProfiler⁶⁵.

Immunoblotting

Cells were lysed in lysis buffer (50 mM Hepes, 150 mM KCl, 1% Triton X-100, 5 mM MgCl₂, pH 7.4) supplemented with a protease/ phosphatase inhibitor cocktail (1mMNa₃VO₄, 10mMNaF, 1 mM PMSF, 10 µg/ml leupeptin, and 10 µg/ml aprotinin). After centrifugation at 12,000 g for 10 min, protein concentrations were measured in the cleared lysates. Equal amounts were resolved by SDS-PAGE followed by immunoblot analysis. Exosome final pellets were resuspended in SDS-PAGE loading dye (50 mM Tris-HCl pH 6.8, 6% glycerol, 2% SDS, 0.2% bromophenol blue), heated briefly at 95°C, resolved by 10% SDS-PAGE gel and transferred onto nitrocellulose membranes (Bio-Rad Laboratories; catalog no. 1620115) using a Bio-Rad Trans-Blot system. Membranes were blocked with 5% skim milk in TBS with Tween-20 for 60 min at RT. Primary antibodies were diluted in blocking buffer and incubated overnight at 4°C. HRP-conjugated secondary antibodies (Bio-Rad Laboratories) diluted in blocking buffer at 1:3000 were incubated for 60 min at RT and developed using EZ-ECL (Biological Industries) or SuperSignal West Femto (Thermo Scientific; catalog no. 34094). Blots were imaged using an ImageQuant LAS 4000 system (GE Healthcare) and quantified using ImageJ software.

Transmission Electron Microscopy

Cells in culture were washed in PBS and fixed for 1h in 2.5% glutaraldehyde in 0.1 M phosphate buffer (PB) at RT. Next, samples were slowly and gently scraped and pelleted in 1.5 ml tubes. Pellets were washed in PB and incubated with 1% OsO₄ for 90 min at 4 °C. Then samples were dehydrated, embedded in Spurr and sectioned using Leica ultramicrotome (Leica Microsystems). Ultrathin sections (50–70 nm) were stained with 2% uranyl acetate for 10 min, a lead-staining solution for 5 min and observed using a transmission electron microscope, JEOL JEM-1010 fitted with a Gatan Orius SC1000 (model 832) digital camera. Multivesicular Endosomes (MVE) and Intraluminal Vesicles (ILV) were identified by morphology and MVE area and ILV number were measured using ImageJ.

Extracellular vesicle isolation

MEF adherent cell cultures were seeded at 3–4 × 10⁶ cells/plate in EV-free complete DMEM and grown for 24–48 h. Cell cultures were centrifuged at 300 g, 5 min to pellet cells, and supernatants were centrifuged again at 2100 g, 20 min to pellet dead cells. After this, filtration through 0.22 µm filter units was performed. Filtered media were ultracentrifuged for 75 min at 100,000 g in SW32Ti rotor to pellet extracellular vesicles. The supernatant was discarded, and the pellet (small EVs) re-suspended in phosphate-buffered saline (PBS) and centrifuged again for 75 min at 100,000 g in 140AT rotor. exosome pellets were frozen and stored at –80°C prior to further processing.

Lipidomic analysis

Targeted quantitative ultra-performance liquid chromatography tandem mass spectrometry was used to accurately quantitate the three geometrical isomers (2,2-, 2,3- and 3,3-) of di-22:6-BMP and di-18:1-BMP in control or treated MEF cells. Lipidomic analyses were conducted by Nextcea, Inc. as previously described⁶⁶. Standard curves were prepared using authentic BMP reference standards. Protein was determined by bicinchoninic acid protein assay. A multiplexed quantitative ultraperformance liquid chromatography tandem mass spectrometry method was used to simultaneously quantitate sample glucosylceramides (GluCer d18:1/16:0, d18:1/18:0, d18:1/22:0,

d18:1/24:0, d18:1/24:1), galactosylceramides (GalCer d18:1/16:0, d18:1/18:0, d18:1/22:0, d18:1/24:0, d18:1/24:1), glucosylsphingosine (GluSph 18:1), and galactosylsphingosine (GalSph 18:1). Standard curves were prepared from related standards using a class-based approach. Internal standards were used for each analyte reported. A SCIEX Triple Quad 7500 mass spectrometer was used in positive electrospray ionization mode for detection (SCIEX, Framingham, MA, USA). Injections were made using a Shimadzu Nexera XR UPLC (ultraperformance liquid chromatography) system (Shimadzu Scientific Instruments, Kyoto, Japan). The instruments were controlled by SCIEX OS 2.0 software.

Metabolic labeling

WT and R1441G LRRK2 mutant MEF cells were treated with 200nM MLI-2 or 300 μ M CBE for 16h. Cells were washed twice with PBS and incubated in serum-free DMEM $\pm$ MLI-2 or CBE for 1h. Next, incubate cells in 0.167mM 13 C-labeled Oleic Acid (OA) and 0.167mM deuterated docosahexaenoic acid (DHA) in serum-free DMEM containing 0.0278mM BSA for 25 min. Cells were washed with serum-free DMEM and chased with serum-free DMEM for short time points (0, 15, 30, 45, 60min) or complete DMEM for long timepoints (8, 24 and 48h). Cells were washed with PBS, trypsinized and collected in cold PBS. Cells were pelleted by centrifugation at 1000rpm for 7min and stored at -80°C prior to further processing.

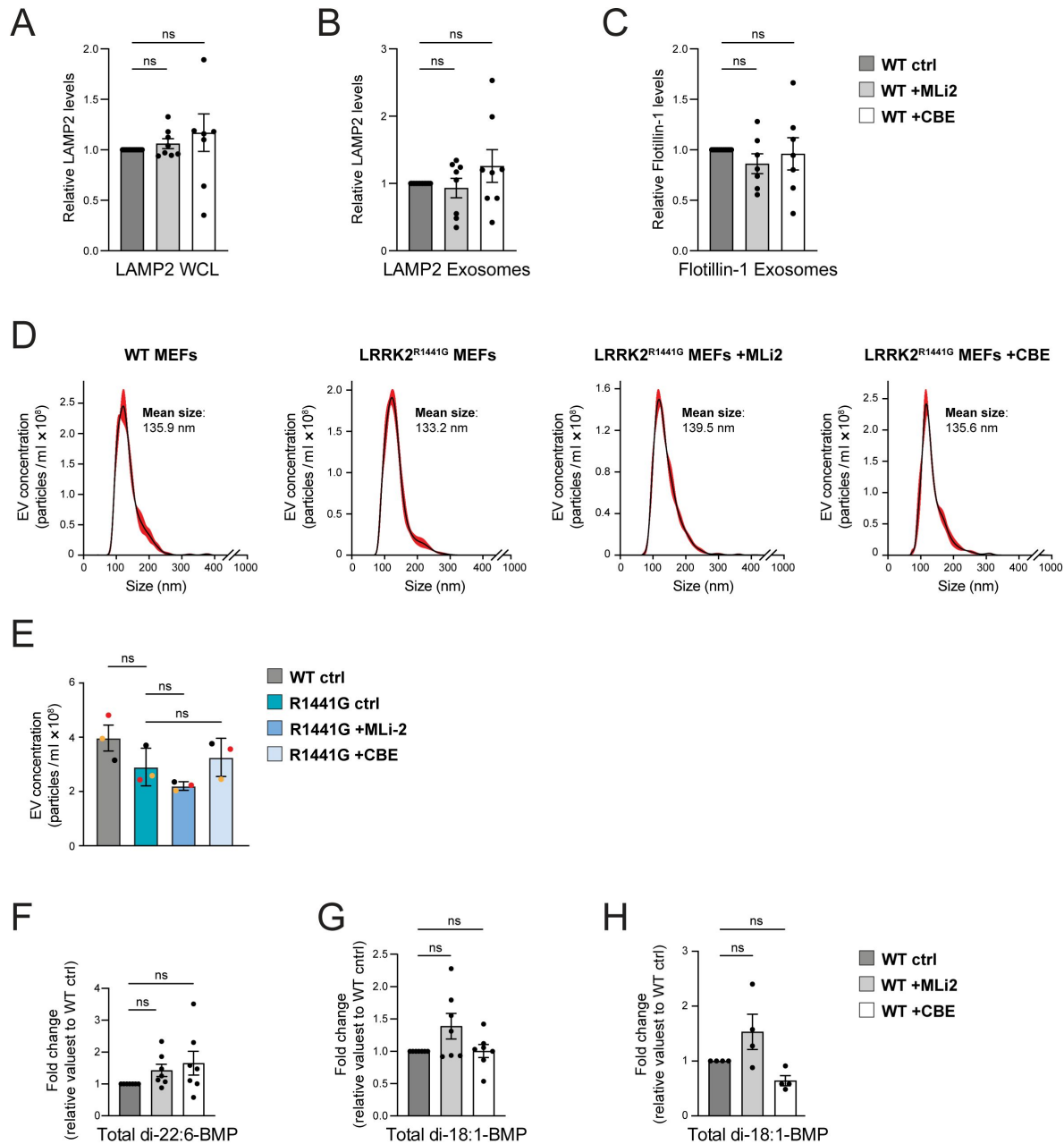
GCcase activity

GCcase activity was measured by using the fluorogenic substrate PFB-FDGlu (Invitrogen, Carlsbad, CA). WT and R1441G LRRK2 mutant MEF cells were treated with 300 μ M CBE for 24h. After this, 150 μ g/ml PFB-FDGlu (Thermo Fisher) was then added to cells followed by incubation for 2h at 37°C . Cells were then washed twice with PBS, trypsinized and fixed in cold 2% paraformaldehyde-PBS for 30 min. Cells were washed twice in PBS before assayed on a FACS analyzer (FACScan; BD) equipped with a 488 nm laser and 530 nm-FITC filter. Data were analyzed using Flowjo software (Tree Star).

Statistics

Results are expressed as the mean $\pm$ SEM. Means were compared using Student's t test when two experimental conditions were independently compared. Unless otherwise noted, statistical significance between multiple comparisons was assessed via one-way or two-way ANOVA with uncorrected Fisher's LSD test using GraphPad Prism version 9.

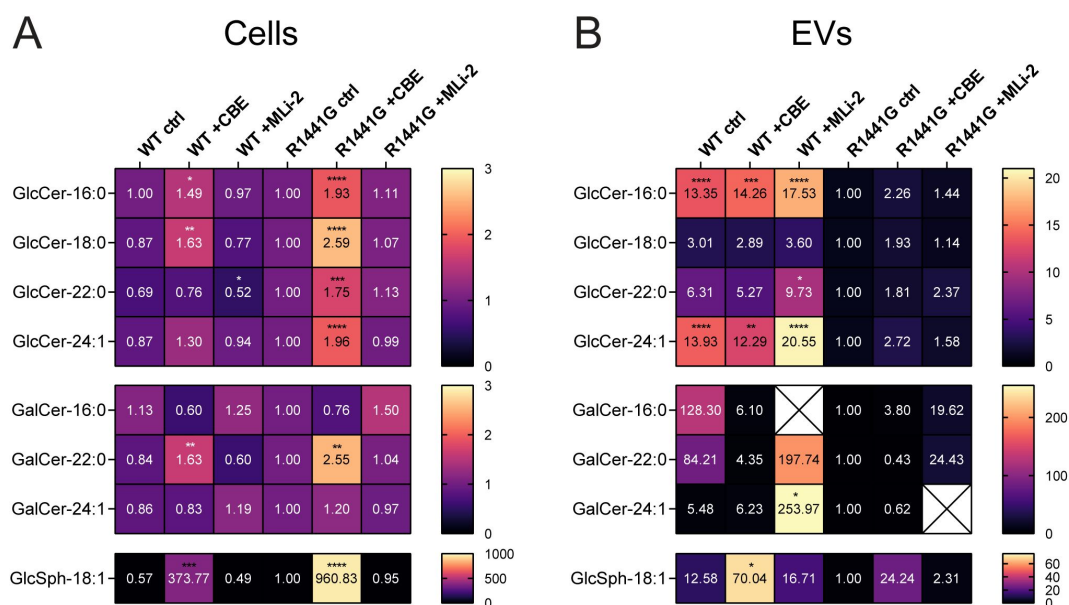
Supplemental figures



Supplemental Figure 1.

Further characterization of MEF-derived EV fractions

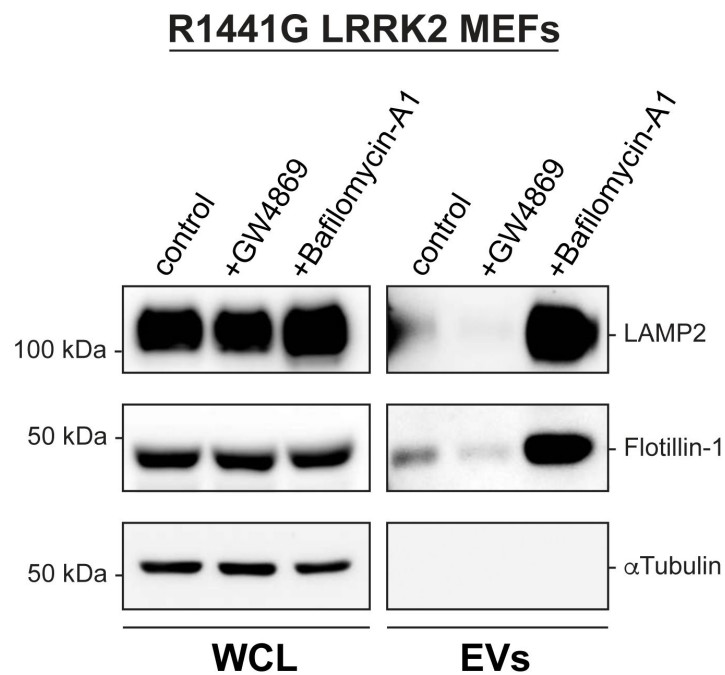
(A-C) No significant differences in EV release between MLI-2/CBE-treated and untreated WT MEF cells. Quantification of LAMP2 and Flotillin-1 levels relative to WT control MEF cells in WCL **(A)** and isolated EVs **(B-C)**. Data from 7-8 independent experiments (mean $\pm$ SEM). Significance determined by ordinary one-way ANOVA, uncorrected Fisher's LSD. **(D)** and **(E)**, Characterization of WT or R1441G LRRK2 MEF-derived purified EVs by Nanoparticle Tracking Analysis (NTA). **(D)** Representative plots of EV size distribution in each indicated condition. **(E)** Yield comparison of MEF-derived EVs from each indicated condition determined by NTA. Each colored dot represents an independent experiment. **(F-H)** No significant differences in EV-associated BMP levels between MLI-2/CBE-treated and untreated WT MEF cells. UPLC-MS/MS measurements of cellular **(F-G)** and EV-associated **(H)** BMP isoforms normalized to protein content from cells treated with 200nM MLI-2 or 300 μ M CBE for 48h or left untreated (ctrl). Data shown as fold change relative to WT control MEF cells. Only BMP isoforms that were detected are shown. Data from 6-7 independent experiments (mean $\pm$ SEM). Significance determined by ordinary two-tailed paired *t* test; ns, not significant.



Supplemental Figure 2.

Quantitative analysis of GCase substrates in MEF cells and EV fractions

(A-B) Lipidomic determination of Glucosylceramide (GlcCer), Galactosylceramide (GalCer) and Glucosylsphingosine (GlcSph) isoforms normalized to protein content from WT and R1441G LRRK2 mutant MEF cells **(A)** and isolated EVs **(B)** treated with 300μM CBE or 200nM MLi-2 for 48h. Heatmap showing values as fold change relative to R1441G LRRK2 mutant MEF cell control. Only GlcCer, GalCer and GlcSph isoforms that were detected are shown. Data from 3-9 independent experiments (mean ± SEM). Significance determined by ordinary two-way ANOVA, uncorrected Fisher's LSD *p<0.05, **p<0.01, ***p<0.001 ****p<0.0001.



Supplemental Figure 3.

Pharmacological modulation of EV-mediated BMP exocytosis in mutant LRRK2 MEF cells

Whole cell lysates (WCL) and isolated EVs from R1441G LRRK2 MEF cells treated with 10μM GW4869 or 10nM bafilomycin-A1 for 24h were analyzed by western blotting. Representative images of LAMP2, Flotillin-1 and α-Tubulin levels are shown. Molecular weight marker mobility is shown in kDa.

Acknowledgements

This research was funded by a grant to K.M and A.L from The Michael J. Fox Foundation for Parkinson's Research (MJFF-019043).

Additional information

Author contributions

K.M.M. and A.L. conceived the project with input from F.H. All experiments were carried out by E.M-S. and A.L. except for mass spectrometric analysis of BMP (F.H), and CLN5 biochemical analysis in human fibroblasts (M.C.). R.F.S., M.E., A.G and M.J.M provided patient-derived fibroblasts. C.E. captured and analyzed transmission electron microscopy images. K.M., A.L., S.R.P. and F.H. oversaw the project. K.M. and A.L. obtained research funding. A.L. wrote and edited the manuscript with critical revision from all co-authors.

Abbreviations

- BMP: bis(monoacylglycerol)phosphate;
- CBE: conduritol b-epoxide;
- DHA: docosahexaenoic acid;
- GCase: glucocerebrosidase;
- ILV: intralumenal vesicle;
- LBPA: Lysobisphosphatidic acid;
- LRRK2: leucine-rich repeat kinase 2;
- PD: Parkinson's disease;
- WT: wild type

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Reviewer #1 (Public review):

Summary:

Even though mutations in LRRK2 and GBA1 (which encodes the protein GCase) increase the risk of developing Parkinson's disease (PD), the specific mechanisms driving neurodegeneration remain unclear. Given their known roles in lysosomal function, the authors investigate how LRRK2 and GCase activity influence the exocytosis of the lysosomal lipid BMP via extracellular vesicles (EVs). They use fibroblasts carrying the PD-associated LRRK2-R1441G mutation and pharmacologically modulate LRRK2 and GCase activity.

Strengths:

The authors examine both proteins at endogenous levels, using MEFs instead of cancer cells. The study's scope is potentially interesting and could yield relevant insights into PD disease mechanisms.

Weaknesses:

Many of the authors' conclusions are overstated and not sufficiently supported by the data. Several statistical errors undermine their claims. Pharmacological treatment is very long, leading to potential off-target effects. Additionally, the authors should be more rigorous when using EV markers.

<https://doi.org/10.7554/eLife.106330.1.sa1>

Reviewer #2 (Public review):

Summary:

In this paper, the authors used MEFs expressing the R1441G mutant of leucine-rich repeat kinase 2 (LRRK2), a mutant associated with the early onset of Parkinson's disease. They report that in these cells LAMP2 fluorescence is higher but BMP fluorescence is lower, MVE size is reduced, and that MVEs contain less ILVs. They also report that LAMP2-positive EVs are increased in mutant cells in a process sensitive to LRRK2 kinase inhibition but are further increased by glucocerebrosidase (GCase) inhibition, and that total di-22:6-BMP and total di-18:1-BMP are increased in mutant LRRK2 MEFs compared to WT cells by mass spectrometry. They also report that LRRK2 kinase inhibition partially restores cellular BMP levels, and that GCase inhibition further increases BMP levels, and that in EVs from the LRRK2 mutant, LRRK2 inhibition decreases BMP while GCase inhibition has the opposite effect. Moreover, they report that the BMP increase is not due to increased BMP synthesis, although the authors observe that CLN5 is increased in LRRK2 mutant cells. Finally, they report that GW4869 decreases EV release and exosomal BMP, while bafilomycin A1 increases EV release. They conclude that LRRK2 regulates BMP levels (in cells) and release (via EVs). They also conclude that the process is modulated by GCase in LRRK2 mutant cells, and that these studies may contribute to the use of BMP-positive EVs as a biomarker for Parkinson's disease and associated treatments.

Strengths:

This is an interesting paper, which provides novel insights into the biogenesis of exosomes with exciting biomedical potential. However, I have comments that authors need to address to clarify some aspects of their study.

Weaknesses:

(1) The intensity of LAMP2 staining is increased significantly in cells expressing the R1441G mutant of LRRK2 when compared to WT cells (Figure 1C). Yet mutant cells contain significantly smaller MVEs with fewer ILVs, and the MVE surface area is reduced (Figure 1D-F). This is quite surprising since LAMP2 is a major component of the limiting membrane of

late endosomes. Are other proteins of endo-lysosomes (eg, LAMP1, CD63, RAB7) or markers (lysotracker) also decreased (see also below)?

(2) LRRK2 has been reported to interact with endolysosomal membranes. Does the R1441G mutant bind LAMP2- and/or BMP-positive membranes? Does the mutant affect endolysosomes?

(3) Immunofluorescence data indicate that BMP is decreased in mutant LRRK2-expressing cells compared to WT (Figure 1A-B), but mass spec data indicate that di-22:6-BMP and di-18:1-BMP are increased (Figure 3). Authors conclude that the BMP pool detected by mass spec in mutant cells is less antibody-accessible than that present in wt cells, or that the anti-BMP antibody is less specific and that it detects other analytes. This is an awkward conclusion, since the IF signal with the antibody is lower (not higher): why would the antibody be less specific? Could it be that the antibody does not see all BMP isoforms equally well? Moreover, the observations that mutant cells contain smaller MVEs (Figure 1D-F) with fewer ILVs are consistent with the IF data and reduced BMP amounts. This needs to be clarified.

Mass spectrometry data are only shown for two BMP species (di-22:6, di-18:1). What are the major BMP isoforms in WT cells? The authors should show the complete analysis for all BMP species if they wish to draw quantitative conclusions about the amounts of BMP in wt and mutant cells. Finally, BMP and PG are isobaric lipids. Fragmentation of BMPs or PGs results in characteristic fingerprints, but the presence of each daughter ion is not absolutely specific for either lipid. This should be clarified, e.g., were BMP and PG separated before mass spec analysis? Was PG affected? The authors should also compare the BMP data with mass spec data obtained with a control lipid, e.g., PC.

(4) It is quite surprising that the amounts of labeled BMP continue to increase for up to 24h after a short 25min pulse with heavy BMP precursors (Figure 4B).

(5) It is argued that upregulation of CLN5 may be due to an overall upregulation of lysosomal enzymes, as LAMP2 levels were also increased (Figure 2A, C, E). Again, this is not consistent with the observed decrease in MVE size and number (Figure 1D-F). As mentioned above, other independent markers of endo-lysosomes should be analyzed (eg, LAMP1, CD63, RAB7), and/or other lysosomal enzymes (e.g. cathepsin. D).

(6) The authors report that the increase in BMP is not due to an increase in BMP synthesis (Figure 4), although they observe a significant increase in CLN5 (Figure 5A) in LRRK2 mutant cells. Some clarification is needed.

(7) Authors observe that both LAMP2 and BMP are decreased in EVs by GW4869 and increased by bafilomycin (Figure 6). Given my comments above on Figure 1, it would also be nice to illustrate/quantify the effects of these compounds on cells by immunofluorescence.

<https://doi.org/10.7554/eLife.106330.1.sa0>